

ENTERPRISE LEARNING PLATFORM WITH SKILL AND CAREER GUIDANCE SYSTEM

(Milestone 4)

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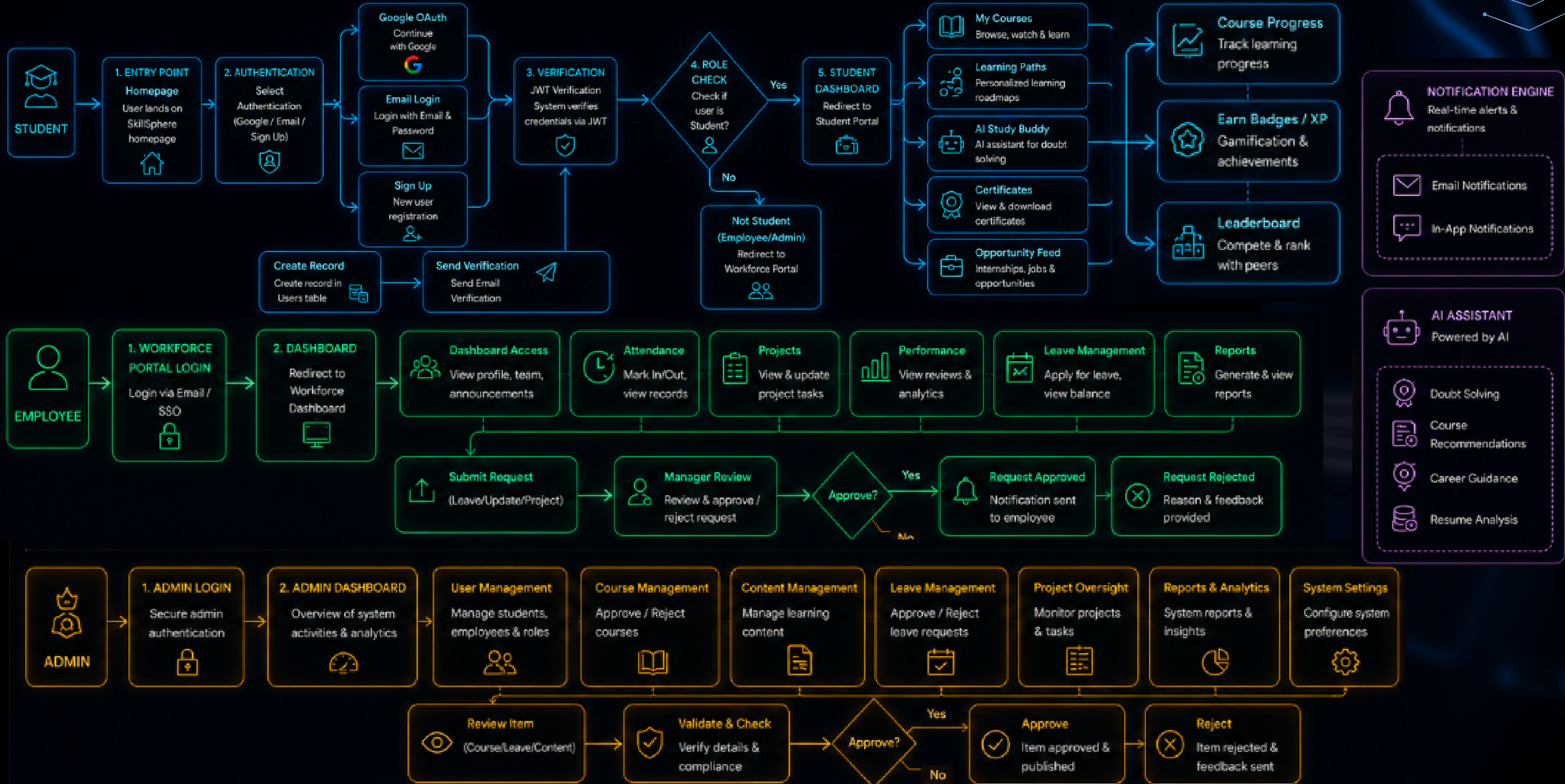
INTRODUCTION

- **Enterprise learning platform is an AI-powered Learning & Workforce Management Platform designed to bridge the gap between skill development and workplace productivity.**
- **Integrates personalized learning, AI-driven recommendations, gamification, and workforce management within a unified platform.**
- **Enables students to follow personalized learning paths, track skills, practice coding, and earn certifications.**
- **Helps organizations manage employees, assign training, monitor performance, and enhance workforce productivity.**
- **Provides real-time analytics and intelligent insights to support continuous learning, performance improvement, and career growth.**

PROBLEM STATEMENT

- **Academic-Industry Skill Mismatch:** Traditional curriculum prioritize theoretical concepts over practical engineering skills, creating a critical gap in job-ready technical talent.
- **Siloed & Fragmented Ecosystem:** Learners endure high cognitive friction by constantly switching across disconnected tools for courseware, coding practice, resume building, and job search.
- **Deficit in Adaptive Mentorship:** Static learning frameworks fail to deliver 24/7 personalized guidance, real-time code debugging, or dynamic career pathing.
- **Recruitment Inefficiencies & Friction:** Employers face prolonged hiring cycles and high screening overhead due to manual resume evaluations lacking verified candidate skill data.

WORKFLOW DIAGRAM



TECH STACK



1. FRONTEND LAYER



React 19
Core Library



JavaScript (ES6+)
Programming Language



Vite
Build Tool



React Router DOM v7
Routing & State



Custom CSS3
Styling (Modern UI, Glassmorphism, Micro-animations)



2. BACKEND LAYER



Java 17
Language & JDK



Spring Boot 3.2.5
Framework (RESTful Web Services)



Spring Security
Security & Role-Based Access Control (RBAC)



Apache Maven
Build Tool



3. DATABASE & STORAGE



MySQL 8.0
Primary Relational Database



Hibernate
ORM Framework (Hibernate & Spring Data JPA)



Cloud Clever Database
Cloud Database Platform



4. AUTHENTICATION & SECURITY



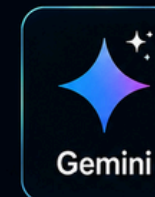
Stateless Auth
JWT (JSON Web Tokens)



Social Login
Google OAuth 2.0 Integration



5. AI & EXTERNAL INTEGRATIONS



Google Gemini AI
gemini-1.5-flash / gemini-2.0-flash
AI engine for intelligent responses and content generation



YouTube API Integration
Video services for embedding, search and playback integration



LOGIN PAGE

.....

Welcome Back!

Let's continue your learning journey

Access your personalized learning space and keep progressing towards your goals.



Student Access Portal

Student

Workforce

Email Address

student@gmail.com

Password

Remember me [Forgot Password?](#)

Log In →

or continue with

 Sign in with Google

Mock Dev Mode
Enter email to bypass Google (STUDENT)

student@skillsphere.com [Bypass](#)

Don't have an account? [Sign up here](#)

.....

Welcome Back!

Let's build, collaborate and create impact.

Access your workspace and tools to manage projects, teams and drive results.



Workforce Access Portal

Student

Workforce

Email Address

workforce@company.com

Password

Remember me [Forgot Password?](#)

Log In →

or continue with

 Sign in with Google

Mock Dev Mode
Enter email to bypass Google (WORKFORCE)

workforce@company.com [Bypass](#)

Don't have an account? [Contact Admin](#)



Admin Portal

Secure access for SkillSphere administrators.

Admin Email

admin@skillsphere.com

Master Password

[Authenticate](#)

[← Return to Homepage](#)

Student Authentication Features

- Secure Student Authentication
- Google OAuth Integration
- Password Recovery System
- Role-Based Authorization

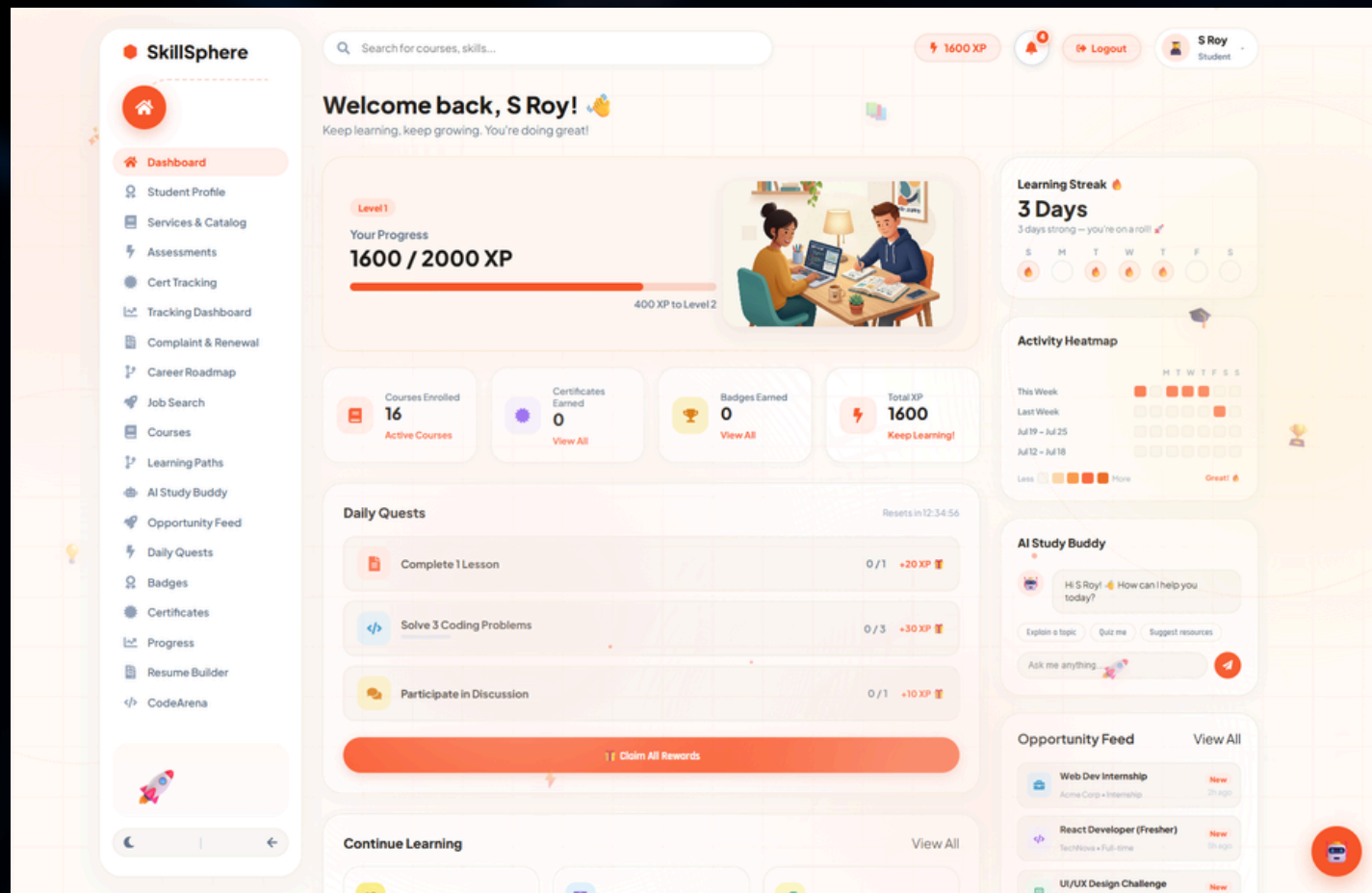
Workforce Authentication Features

- Secure Employee Login
- Organization Email Verification
- Role-Based Access Control

Authentication

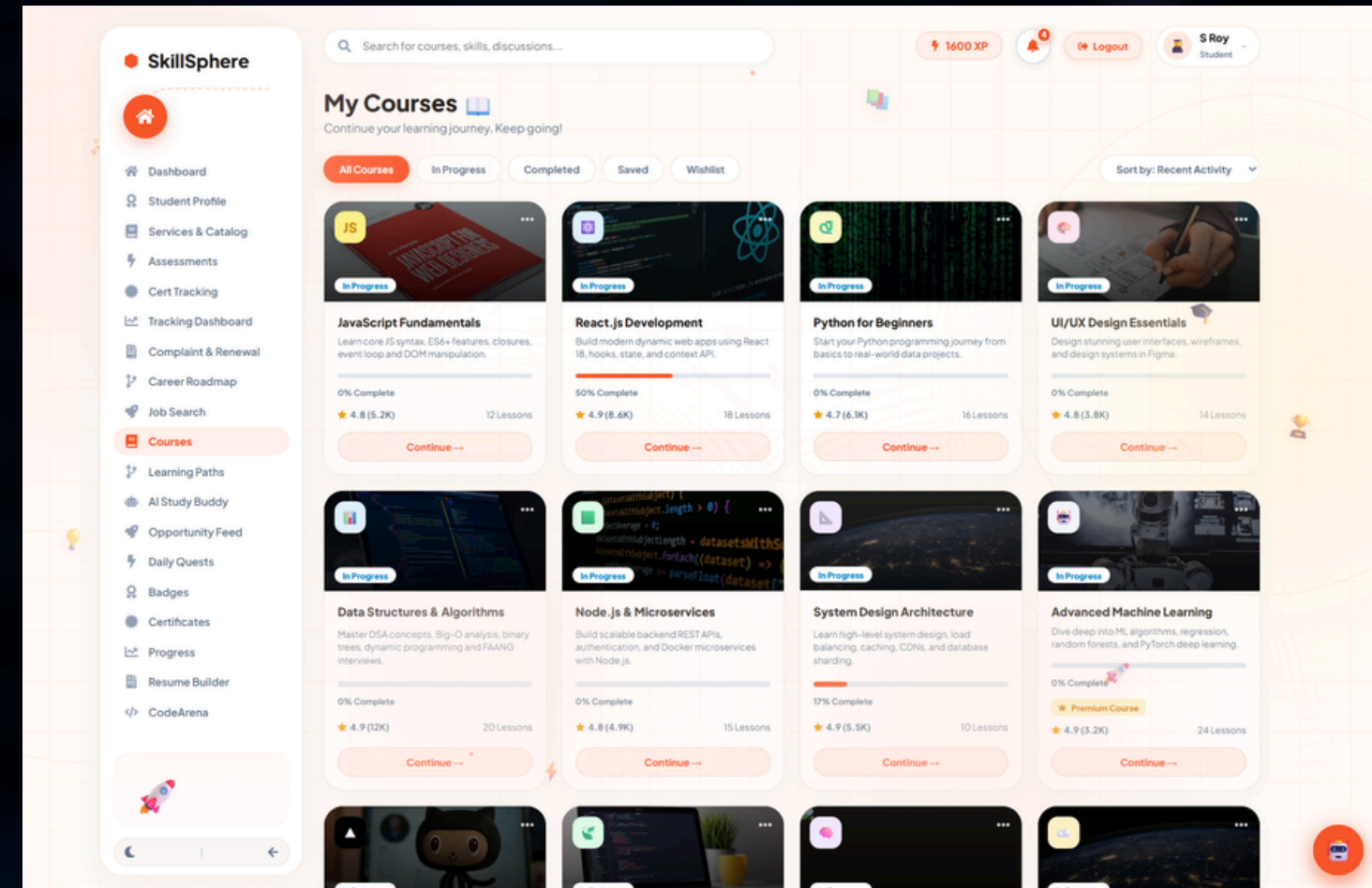
- Secure Administrator Login
- Multi-Level Role Authorization
- JWT Token Validation
- Audit Logging & Security Control

STUDENT DASHBOARD



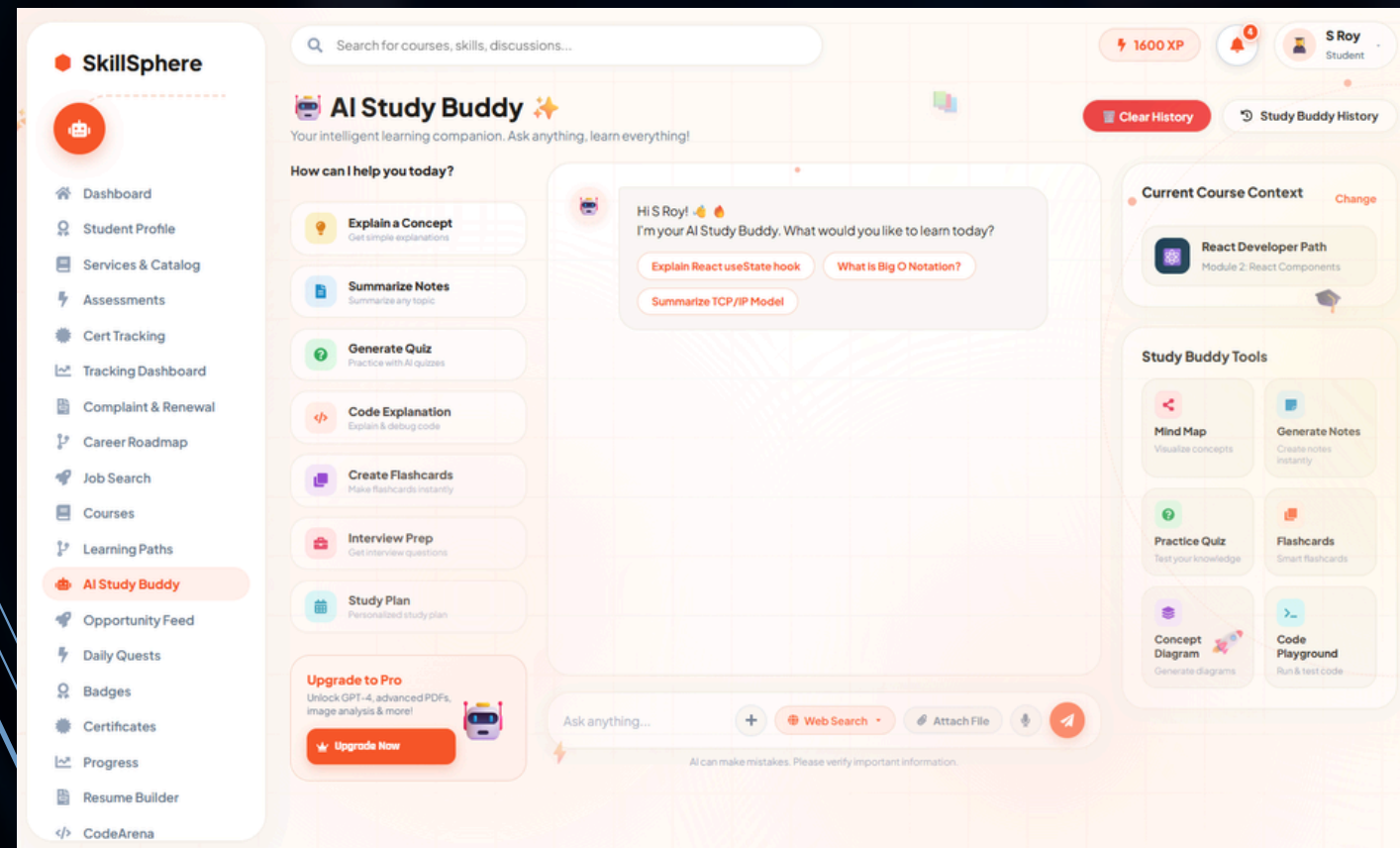
STUDENT DASHBOARD

Overview of learning progree, XP, streak



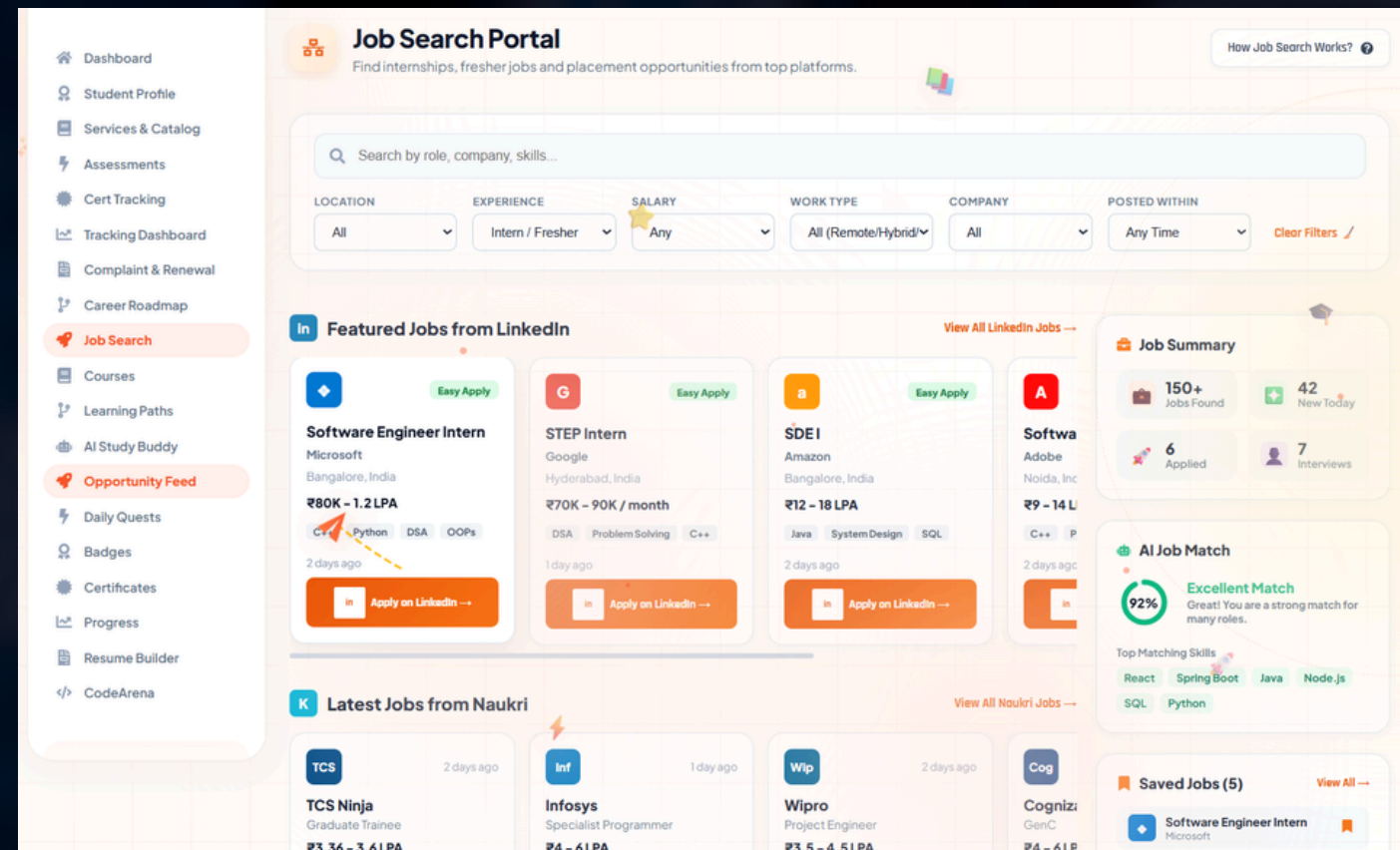
MY COURSES

Track all enrolled courses



AI STUDY BUDDY

Get instant help, quizzes and study support from AI



OPPORTUNITY FEED

Discover job , internships and placemnet opportunities

WORKFORCE & ADMIN DASHBOARD

WORKFORCE DASHBOARD

ADMIN DASHBOARD

Enterprise Learning Platform

Overview

Assessments

My Certificates

Promotion Flowchart

My Leaves & Attendance

AI Career Coach

Sign Out

Workforce Employee Portal

DarkEmployee Demo (Employee)

Welcome back, Employee Demo! 🙌
Track your assignments, learning targets, and skill progress in one single place.

Assessments Taken
0 / 4
Verified certifications

My Certificates
0 Earned
Direct download PDF

Promotion Target
Lv. 3
Senior Engineer Track

My Leaves Filled
0 Requests
Synced with HR

My Active Team Tasks

Refactor Auth Token Verification
Move from local session state to global JWT middleware verification.
High Priority

Implement Vitest Suite
Write unit tests covering routing redirection and context initialization.
Medium Priority

Leave Balance

Casual Leaves
8 / 12 Days Left

Sick Leaves
6 / 8 Days Left

Enterprise Learning Platform

Dashboard

Student Management

Workforce Management

Courses (Admin)

Course Approvals

Leave Approvals

Service Bookings

Certifications

Settings

Manage with Ease
Control enrollment, courses, and verify certifications in one unified portal.

Logout Admin

Dashboard

Welcome back, Admin!
Here's what's happening on your platform today.

Aug 12, 2026

TOTAL STUDENTS
36 +12.5%
from last month

WORKFORCE
20 +8.4%
from last month

ACTIVE COURSES
12 +5.3%
from last month

CERTIFICATES
5 +15.7%
from last month

PENDING APPROVALS
0
All time total

Recent Student Registrations

View All

S Roy
sroy@gmail.com
Jul 22, 2026

Student Demo
student@skillsphere.com
Jul 22, 2026

rema seni
remaseni123@gmail.com
Jul 23, 2026

Test User
testuser_unique_123@gmail.com
Jul 23, 2026

stu12
stu1@gmail.com
Jul 23, 2026

Recent Workforce Registrations

View All

Employee Demo
employee@skillsphere.com (EMPLOYEE)
Jul 22, 2026

Manager Demo
manager@company.com (MANAGER)
Jul 22, 2026

employ1new
employ1@gmail.com (EMPLOYEE)
Jul 22, 2026

New Employee
newemployee@skillsphere.com (EMPLOYEE)
Jul 22, 2026

employ1
employ1@skillsphere.com (EMPLOYEE)
Jul 22, 2026

Quick Actions

Enterprise Learning Platform

Overview

Assessments

My Certificates

Promotion Flowchart

My Leaves & Attendance

AI Career Coach

Sign Out

Workforce Employee Portal

DarkEmployee Demo (Employee)

Career Path & Promotion Roadmap
Select a specialized track to view the promotional flowchart. Checkpoints automatically unlock as you pass the corresponding verified assessments.

Full-Stack Developer Path

DevOps Architect Path

AI/ML Engineer Path

Lvl1 Associate Engineer
Requires: JavaScript Basics Check passed.
Unlocked

Lvl2 Mid-Level Specialist
Requires: React Hooks Proficiency assessment passed.
Locked

Lvl3 Team Tech Lead
Requires: Spring Boot Microservices & JPA assessment passed.
Locked

Lvl4 Principal Solutions Architect
Requires: AWS Cloud & Infrastructure Essentials passed.
Locked

Enterprise Learning Platform

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Course Approvals

Leave Approvals

Service Bookings

Certifications

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Manage with Ease
Control enrollment, courses, and verify certifications in one unified portal.

Logout Admin

Workforce Leave Approvals

Review, approve, or reject leave requests submitted by workforce employees and managers in real-time.

Sync Requests

Pending Leave Requests 2 pending

EMPLOYEE	LEAVE TYPE	DATES & DURATION	REASON	ACTIONS
Alex Vance alex@skillsphere.com • Engineering	Sick Leave	2026-08-05 to 2026-08-07 3 days	High fever and doctor advised rest	Approve Reject
David Miller david@skillsphere.com • Marketing	Paid Time Off	2026-08-10 to 2026-08-12 3 days	Personal annual trip	Approve Reject

Processed Leave History

EMPLOYEE	LEAVE TYPE	DATES	STATUS
Riya Sharma riya@skillsphere.com	Casual Leave	2026-08-02 to 2026-08-03 (2d)	Approved

CONCLUSION

- **Unified AI Ecosystem:** Integrates learning, skill building, career growth, and workforce management on a single platform.
- **Comprehensive Features:** Combines personalized learning, AI assistance, coding practice, career opportunities, and enterprise workforce solutions.
- **Multi-Audience Impact:** Empowers learners to excel, professionals to grow, and organizations to build future-ready teams.
- **Future-Proof Architecture:** Secure, scalable, and designed to adapt to next-generation technologies.
- **Core Vision:** Shaping the future of learning and work.

THANK YOU